(22) Problems:-

Ex (1) The population of town is doubled for every 6 years from 2000 to 2024. Then % change in population in this duration?

Ans:- Method:1:- Let assume Initial pop^n as 'x'

$$2000 \xrightarrow[+6y]{\times 2} 2006 \xrightarrow[+6y]{\times 2} 2012 \xrightarrow[+6y]{\times 2} 2018 \xrightarrow[+6y]{\times 2} 2024$$
$$x \qquad 2x \qquad 4x \qquad 8x \qquad 16x$$
(Initial) [final]

$$\%\text{change} = \frac{f-1}{1} \times 100 = \frac{16x - x}{x} \times 100\% = 1500\% \text{ ↑ed}$$

Method 2:- Initial pop^n as 100%

$$2000 \xrightarrow{\times 2} 2006 \xrightarrow{\times 2} 2012 \xrightarrow{\times 2} 2018 \xrightarrow{\times 2} 2024$$
$$100\% \qquad 200\% \qquad 400\% \qquad 800\% \qquad 1600\%$$

$\%\text{change} = 1600\% - 100\% = 1500\%$ [directly]

Method 3:- Initial pop^n as '1'

$$2000 \xrightarrow{\times 2} 2006 \xrightarrow{\times 2} 2012 \xrightarrow{\times 2} 2018 \xrightarrow{\times 2} 2024$$
$$1 \qquad 2 \qquad 4 \qquad 8 \qquad 16$$

$$\%\text{change} = \frac{f-1}{1} \times 100 = \frac{(16-1)}{1} \times 100 = 1500\%$$

Method 4 (or) Formula:-

In the above step/method we can write $16 = 2^4$

$$\%\text{change} = (2^4 - 1)\,100 = (\text{No. of times}^{\frac{\text{duration}}{\text{condition}}} - 1)\,100$$
$$= (2^{24/6} - 1)\,100$$

$$\boxed{\%\text{change} = \left[\text{No. of times}^{\frac{\text{Duration}}{\text{Condition}}} - 1\right] 100\ \%}$$

(2) The pop^n of town is ↑ed by 10% in 1st year, 20% in 2nd and 30% in 3rd year on initial pop^n 25000. The pop^n after 3 yrs is?

Ans:- M1:-

25000 → 100%
+ 2500 — 10% of 25k
27500
5500 — 20% of 27500

33000
9900 → 30% of 33k
42900

(or) IM-2:-

after 1 year = (100% + 10%) of 25k

= 110% of 25k = 1.1 × 25k = 27500

after 2 yrs = (1 + 0.2) 1.1 × 25k = 1.2 × 1.1 × 25k = 33000

after 3 yrs = (1 + 0.3) 1.2 × 1.1 × 25k = 42900

Conclusion / formula:-

↳ If popn is ↑ed x% every year for n years on Present popn. The the resultant popn after n years is

$\left(1+\frac{x}{100}\right)\left(1+\frac{x}{100}\right)$ ..... n times × Present Popn

$= \left(1+\frac{x}{100}\right)^n P.P$

↳ If popn is ↑ed x% after 1st, y% after 2nd, 3% after 3rd... on Present popn. the popn after n yrs is.

$\left(1+\frac{x}{100}\right)\left(1+\frac{y}{100}\right)\left(1+\frac{3}{100}\right)$ ... n times × $P.P^n$.

↳ If popn ↑ed / ↓ed one after the other for n years then the popn after n years

$\left(1\pm\frac{x}{100}\right)\left(1\pm\frac{y}{100}\right)\left(1\pm\frac{3}{100}\right)$ ..... n times × P.Popn.

- ↑ → +ve sign | ↓ → −ve sign

↳ If popn ↑ed by / ↓ed by x% every year for 'n' years. The resultant popn after 'n' years is.

$\left(1\pm\frac{x}{100}\right)^n . P.Popn$

↳ All the above cases are if the problem is asked to find out future popn. [for after 'n' years].

Ex:3:- The pop^n of a town $\uparrow$ed for 3 yrs as 1st year $\to$ 10%, 2nd year $\to$ 20%, 3rd year $\to$ 30% and changed to 17160. The pop^n 3 yrs ago / Initial population?

Ans:- Assume Initial pop^n as 'x'

after 1 yr $\to x \times 1.1$

after 2 yrs $\to x \times 1.1 \times 1.2$

after 3 yrs $\to x \times 1.1 \times 1.2 \times 1.3 = 17160$ = present pop^n

Pop^n 3 years ago = Initial pop^n = $x = \frac{17160}{1.1 \times 1.2 \times 1.3}$

$x = \frac{17160}{1.716} = 10000$

↳ From the above examples

(i) to find future pop^n after 'n' years, the present population is multiplied with respectable factor. [For ↑/↓]

(ii) ~~Every year~~ to find past population 'n' years ago, the present population is devided with respectable factors [↑/↓]

↳ Conclusions:-

(i) Every year the pop^n is ↑ed/↓ed by x%. for 'n' years, the population becomes y. then pop^n n years ago to y is.

Pop^n 'n' years ago = $\frac{\text{Present Pop}^n}{(1 \pm \frac{x}{100})(1 \pm \frac{x}{100}) \cdots n \text{ times}} = \frac{y}{(1 \pm \frac{x}{100})^n}$

[✡ If ↑ed '+' sign, ↓ed '−' sign]

(ii) 1st yr pop^n ↑ed/↓ed by a%, 2nd year ↑ed/↓ed by b%, ... nth year ↑ed/↓ed by x% and becomes y. The pop^n 'n' yrs ago is

Population 'n' yrs ago = $\frac{\text{Present Population}}{(1 \pm \frac{a}{100})(1 \pm \frac{b}{100}) \cdots (1 \pm \frac{x}{100})}$

$= \frac{y}{(1 \pm \frac{a}{100})(1 \pm \frac{b}{100}) \cdots (1 \pm \frac{x}{100})}$ ← n year ↑/↓ %

[If ↑ed '+' sign, ↓ed '−' sign]